Hands-On Activity 10 | Explaining Visually to Influence with Data

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Course Code and Title : CPE 031-CPE21S1 - Visualizations and Data Analysis
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1. Objectives:

This activity aims to help students demonstrate how to influence and communicate effectively with data by identifying audience needs, crafting a clear message, and telling a story visually through charts.

2. Intended Learning Outcomes (ILOs):

By the end of this activity, students should be able to:

1. Analyze the audience to understand their level of data knowledge and needs.
2. Evaluate how well a message and visualization communicate insights.
3. Create a clear and meaningful data story using Python visualization tools.

3. Discussions:

Data storytelling goes beyond showing charts it's about influencing decisions and sparking understanding.

Know Your Audience: Decision-makers may prefer clear summaries, while analysts like detailed data.

Know Your Message: Focus on the why it matters part of the story.

Tell a Story: Use preattentive attributes color, size, and position to guide attention.

Structure Your Story: Like Aristotle's rhetorical triangle, combine logic (logos), emotion (pathos), and credibility (ethos).

4. Procedures:

Step 1. Import Libraries and Create Sample Data

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

# Sample sales dataset
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun"],
    "Sales": [200, 250, 180, 300, 280, 350],
    "Marketing_Spend": [50, 60, 45, 70, 65, 80]
}
df = pd.DataFrame(data)
```

```
12  df
```

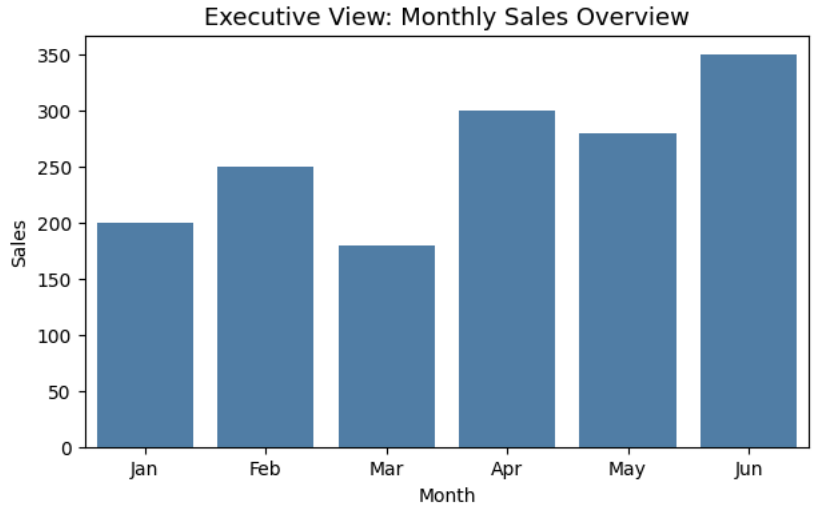
	Month	Sales	Marketing_Spend
0	Jan	200	50
1	Feb	250	60
2	Mar	180	45
3	Apr	300	70
4	May	280	65
5	Jun	350	80

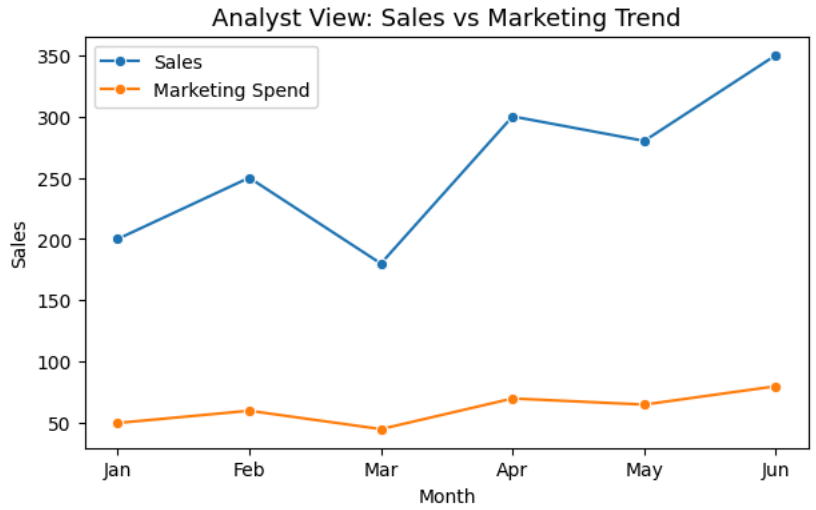
Next steps: Generate code with df | New interactive sheet

Step 2. Know Your Audience — Compare Chart Styles

```
# Executive audience: prefer clarity (simple bar)
plt.figure(figsize=(7,4))
sns.barplot(x="Month", y="Sales", data=df, color="steelblue")
plt.title("Executive View: Monthly Sales Overview", fontsize=13)
plt.show()

# Analyst audience: prefer detailed trends (line chart)
plt.figure(figsize=(7,4))
sns.lineplot(x="Month", y="Sales", data=df, marker="o", label="Sales")
sns.lineplot(x="Month", y="Marketing_Spend", data=df, marker="o", label="Marketing Spend")
plt.title("Analyst View: Sales vs Marketing Trend", fontsize=13)
plt.legend()
plt.show()
```



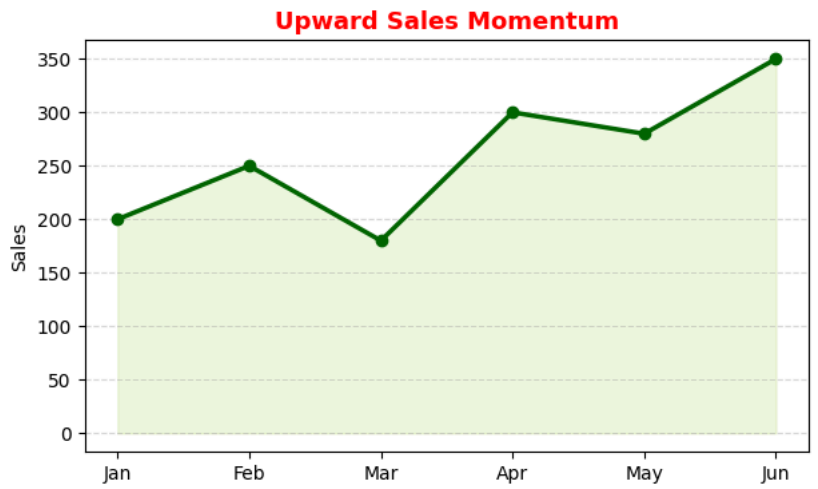


Task 1: Create one additional visualization suited for a different audience (e.g., a customer, investor, or teammate).

```
#Code Here
plt.figure(figsize=(7, 4))
plt.plot(
    df["Month"],
    df["Sales"],
    color="darkgreen",
    marker="o",
    linewidth=2.5,
    label="Monthly Revenue",
)

plt.fill_between(df["Month"], df["Sales"], color="yellowgreen", alpha=0.15)
```

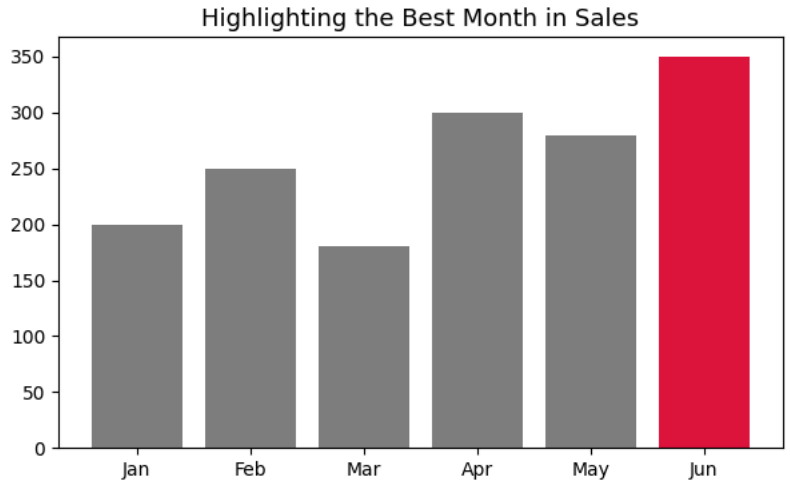
```
plt.title("Upward Sales Momentum",fontsize=13,color="red",fontweight="bold")
plt.ylabel("Sales")#for the investors probably is much better
plt.grid(axis="y", linestyle="--", alpha=0.5)
plt.show()
```



Step 3. Know Your Message — Highlight What Matters

```
# Highlight the highest sales month (preattentive color cue)
highlight = df["Sales"].idxmax()

colors = ["gray" if i != highlight else "crimson" for i in range(len(df))]
plt.figure(figsize=(7,4))
plt.bar(df["Month"], df["Sales"], color=colors)
plt.title("Highlighting the Best Month in Sales", fontsize=13)
plt.show()
```

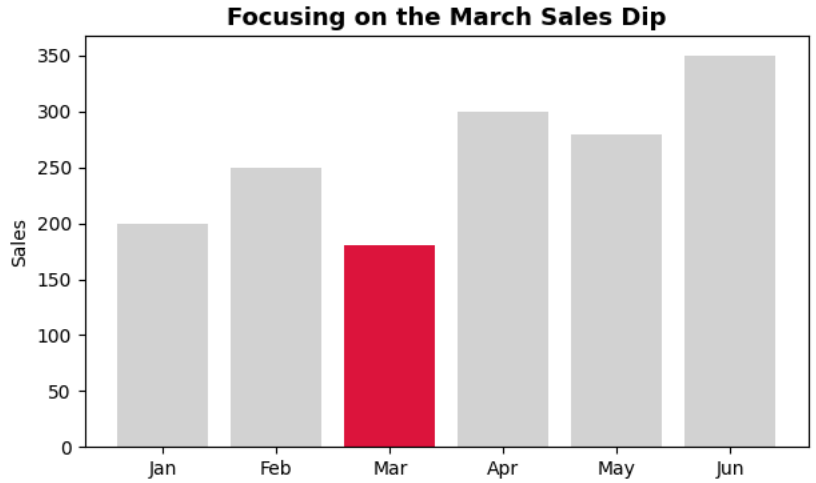


Task 2: Use color or annotations to emphasize a key insight (e.g., lowest point, sudden increase).

```
#Code Here
lowest_idx = df["Sales"].idxmin()
colors = ["crimson" if i == lowest_idx else "lightgray" for i in range(len(df))]

plt.figure(figsize=(7, 4))
bars = plt.bar(df["Month"], df["Sales"], color=colors)
plt.title("March Sales", fontsize=13, fontweight="bold")

plt.ylabel("Sales")
plt.show()
```

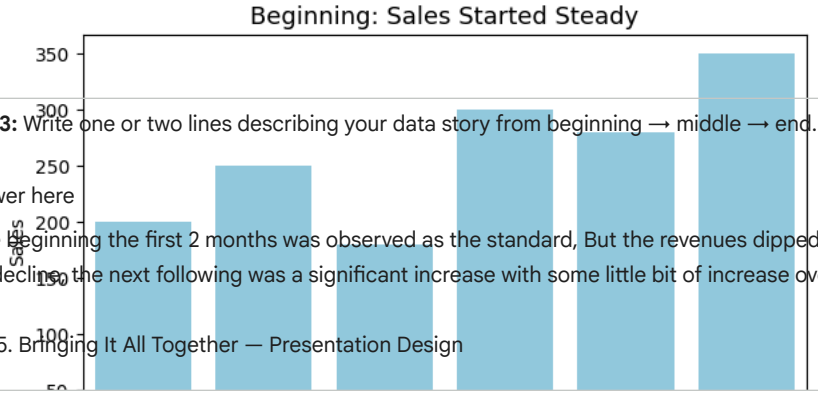


Step 4. Storytelling with Charts — Beginning, Middle, End

```
# Beginning: Context
plt.figure(figsize=(7,4))
sns.barplot(x="Month", y="Sales", data=df, color="skyblue")
plt.title("Beginning: Sales Started Steady", fontsize=13)
plt.show()

# Middle: Change or Challenge
plt.figure(figsize=(7,4))
sns.lineplot(x="Month", y="Sales", data=df, color="orange", marker="o")
plt.title("Middle: A Drop in March Due to Low Marketing", fontsize=13)
plt.show()

# End: Resolution
plt.figure(figsize=(7,4))
sns.barplot(x="Month", y="Sales", data=df, color="limegreen")
plt.title("End: Recovery and Growth After March", fontsize=13)
plt.show()
```



Task 3: Write one or two lines describing your data story from beginning → middle → end.

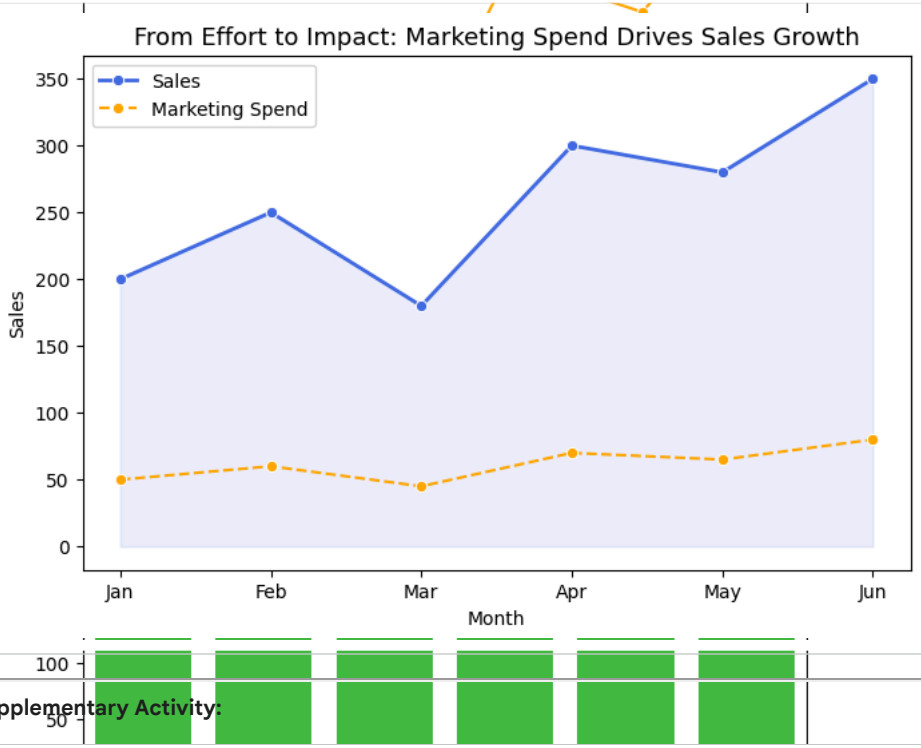
*Answer here

In the beginning the first 2 months was observed as the standard, But the revenues dipped quite a big amount in the next month, but after that decline, the next following was a significant increase with some little bit of increase over time.

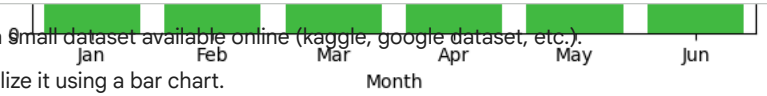
Step 5. Bringing It All Together — Presentation Design

```
# Simple combined story chart
plt.figure(figsize=(8,5))
sns.lineplot(x="Month", y="Sales", data=df, marker="o", linewidth=2, color="royalblue", label="Sales")
sns.lineplot(x="Month", y="Marketing_Spend", data=df, marker="o", linestyle="--", color="orange", label="Marketing Spend")

plt.fill_between(df["Month"], df["Sales"], alpha=0.1, color="royalblue")
plt.title("From Effort to Impact: Marketing Spend Drives Sales Growth", fontsize=13)
plt.legend()
plt.show()
```



5. Supplementary Activity:



1. Use a small dataset available online (kaggle, google dataset, etc.).
2. Visualize it using a bar chart.
3. Highlight one important data point (e.g., the highest or lowest value) using a distinct color.
4. Add a title and annotation that tell the story behind the data.

```
#CODE HERE

import matplotlib.pyplot as plt
import pandas as pd

# Load dataset
df = pd.read_excel('phl_population-projections_admin3_midyear2020_2025.xlsx')

# Aggregate population growth by region
summary = df.groupby('Region')[['July2020', 'July2025']].sum()
summary['Growth_Rate'] = ((summary['July2025'] - summary['July2020'])/ summary['July2020']) * 100

summary = summary.sort_values(by='Growth_Rate', ascending=True)

top_2_regions = ['Autonomous Region in Muslim Mindanao (ARMM)','Region IV-A (CALABARZON)']
lowest_region =summary.index[0]

colors = []
for reg in summary.index:
    if reg in top_2_regions:
        colors.append('crimson')
    elif reg == lowest_region:
        colors.append('darkblue')
    else:
        colors.append('lightgray')

# Creating the horizontal bar chart
plt.figure(figsize=(10, 6))
bars = plt.barh(summary.index, summary['Growth_Rate'], color=colors)

# Storytelling Title and Labels
plt.title('Philippine Regional Population Growth (2020–2025)',fontsize=13,fontweight='bold',pad=15)

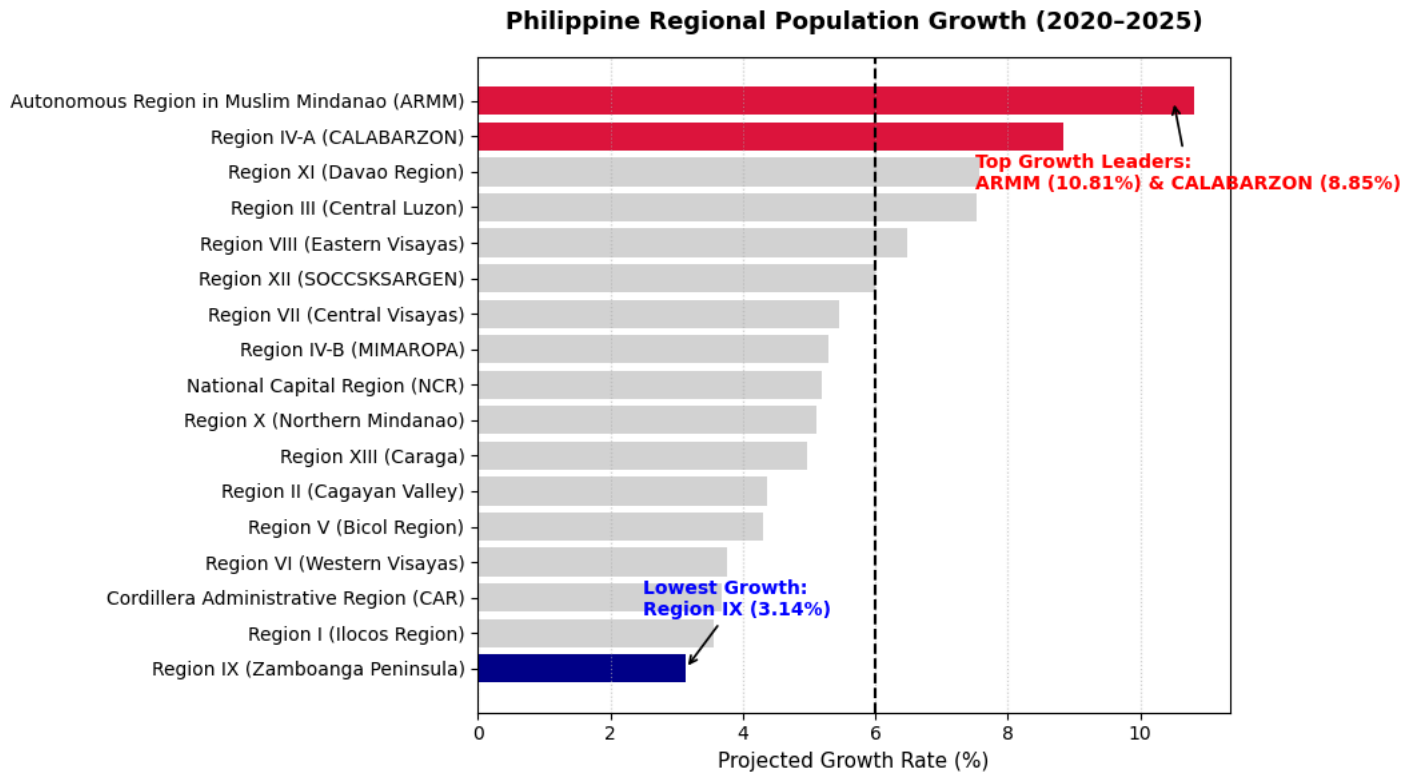
plt.xlabel('Projected Growth Rate (%)', fontsize=11)
plt.axvline(x=6,color='black',linestyle='--',linewidth=1.5,label='National Average (6%)')

#Callout Annotations(thanks online sources)
# High-growth annotation (Top)
plt.annotate(
    'Top Growth Leaders:\nARMM (10.81%) & CALABARZON (8.85%)',
    xy=(10.5, 16),
    xytext=(7.5, 13.5),
    arrowprops=dict(facecolor='red', arrowstyle='->', lw=1.2),
    fontweight='bold',
    color='red',
)

# Low-growth annotation (Bottom)
plt.annotate(
```

```
    f'Lowest Growth:\nRegion IX ({summary.loc[lowest_region, "Growth_Rate"]:.2f}%)',
    xy=(summary.loc[lowest_region, 'Growth_Rate'], 0),
    xytext=(2.5, 1.5),
    arrowprops=dict(facecolor='red', arrowstyle='->', lw=1.2),
    fontweight='bold',
    color='blue',
)

plt.grid(axis='x', linestyle=':', alpha=0.6)
plt.tight_layout()
plt.show()
```



6. Conclusion/Learnings/Analysis:

Close

*Type it here

From this activity, I was able to understand how to be able to give an
effective story telling using coding, like make the general data first, maybe

```
to overwhelm or to make the reader think of what does it represent, and maybe
even select the few who truly have the general or even the idea about the
topic.

I also was able to make a flow, like if my audience was an investor, It would
be better to first use a flow where the data is positive, and if there are
negatives, I could probably like circumvent it by, not manipulating the data,
but to phrase it that the negative is only temporary and would not hinder or
affect the project in the long run, and that maybe even taking this negative,
could make the product bounce back greater, like in the procedure activity.

Also, using online sources, I was able to know about the annotations, but it is
hard to use because of the perfection that you would need to make it be at the
perfect ending, but if your data is even or does not have any decimals, it is
quite easy to use.

Overall, this activity made me able to properly have the foundation for data
story telling and the "coerce" the audience in a data focus kind of way.
```

*Type it here

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